7 **(a)** Fig. 7.1 shows a plan view of a room. There is a plane mirror on one wall and a picture across the whole of wall AB.

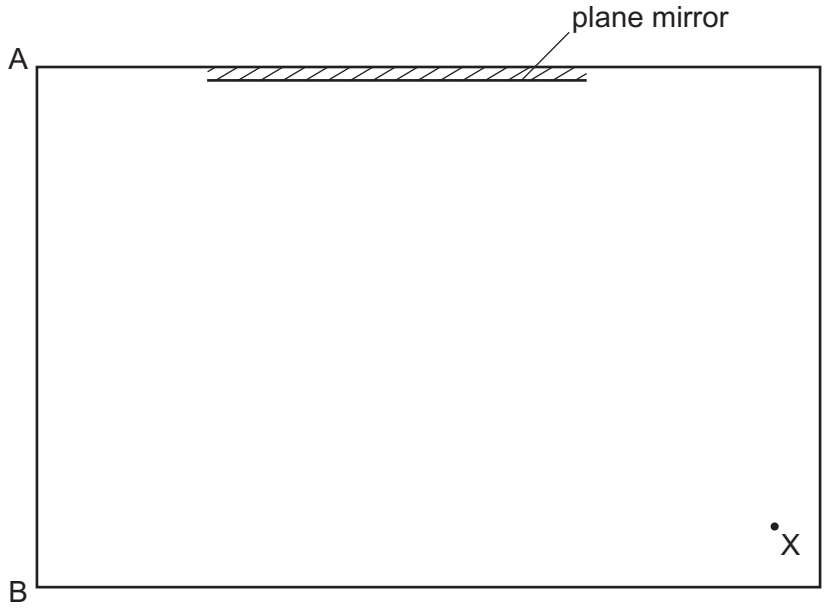


Fig. 7.1 (plan view)

A person is standing at point X and is looking at the mirror. The person cannot see all of the picture on wall AB reflected in the mirror.

There is a point P on wall AB which is the closest point to A that the person can see reflected in the mirror.

On Fig. 7.1, draw a reflected ray and an incident ray to show the position of the point P. [2]

(b) State **two** properties of the image formed by the mirror.

1. .....

2. .....
[2]

(c) Visible light is an electromagnetic wave.

State the name of **one** region of the electromagnetic spectrum in which the waves have:

(i) shorter wavelengths than visible light

..... [1]

(ii) longer wavelengths than visible light.

..... [1]

[Total: 6]

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